Note 16 - Derivatives and the Shape of the Graph

Maximum and Minimum Values

- A graph can be described with its minimums and maximums, defined as follows:
  - $f(c)$ is the **absolute maximum value** of f on D if $f(c) \geq f(x)$ for all $x \in D$.
  - $f(c)$ is the **absolute minimum value** of f on D if $f(c) \leq f(x)$ for all $x \in D$.
    Note: c is a number in the domain D of a function f.
  - $f(c)$ is a **local maximum value** of f if $f(c) \geq f(x)$ when x is *near* c.
  - $f(c)$ is a **local minimum value** of f if $f(c) \leq f(x)$ when x is *near* c.
    *Note: **Near** meaning true on some interval containing c.*
- These definitions lead to the following properties and theorems:
  - A local min/max *cannot* be an endpoint of the domain.
  - An absolute min/max is *either* a local min/max *or* an endpoint.
  - **The Extreme Value Theorem**:
    If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value, $f(c)$, and an absolute minimum value, $f(d)$, at some numbers $c, d \in [a, b]$.
  - **Fermat's Theorem**: If f has a local maximum or minimum at c, and if $f'(c)$ exists, then $f'(c) = 0$.
- **Critical number**: A number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Example: Find the critical numbers of $f(x) = x^{3/5}(4 - x)$.
Procedure: Find the derivative, $f'(x)$. Then, set $f'(x) = 0$ and find the values of x where $f'(x)$ does not exist.

$$f'(x) = x^{3/5}(-1) + \frac{3}{5}x^{-2/5}(4 - x) \qquad \text{Use the product rule to derive the function.}$$

$$= -x^{3/5} + \frac{3(4 - x)}{5x^{2/5}} \qquad \text{Simplify.}$$

$$= \frac{-5x + 3(4 - x)}{5x^{2/5}} \qquad \text{Multiply the first term by } \frac{5x^{2/5}}{5x^{2/5}} \text{ to have a common denominator.}$$

$$= \frac{-8x + 12}{5x^{2/5}} \qquad \text{Simplify the numerator.}$$

Now that we have $f'(x)$, check the two cases for critical numbers:

- Case 1: $f'(c) = 0$

  $$\frac{-8x + 12}{5x^{2/5}} = 0$$

  $$-8x + 12 = 0 \implies x = \frac{3}{2}$$

- Case 2: $f'(c)$ DNE
  Recall: A rational function is undefined when the denominator is 0.

  $$5x^{2/5} = 0 \implies x = 0$$

The critical numbers of f are $x = 0, \frac{3}{2}$.

- If f has a local maximum or minimum at c, then c is a critical number of f.

The Closed Interval Method

To find the absolute minimum and maximum values of a continuous function f on a closed interval $[a, b]$, use the **Closed Interval Method**.

Procedure:

1. Find the values of f at the critical numbers of f in $[a, b]$.
2. Find the values of f at the endpoints of the intervals.
3. The largest of the values from steps (1) and (2) is the absolute maximum. The smallest is the absolute minimum.

Example: Find the absolute minimum and maximum of $f(x) = x^3 - 3x^2 + 1$ on the interval $[-\frac{1}{2}, 4]$.

Step 1: Find the critical numbers of f:

$$\begin{aligned} f'(x) &= 3x^2 - 6x && \text{Use the power rule to derive the function.} \\ &= 3x(x-2) && \text{Factor to solve for } x. \\ \implies x &= 0, 2 && f \text{ has critical numbers at } x = 0 \text{ and } x = 2. \end{aligned}$$

And evaluate f at the critical numbers:
Note: Make sure to plug critical numbers into the original $f(x)$, not the $f'(x)$ you calculated above!

- At $x = 0$:

$$\begin{aligned} f(0) &= (0)^3 - 3(0)^2 + 1 \\ &= 0 - 0 + 1 \\ &= 1 \end{aligned}$$

- At $x = 2$:

$$\begin{aligned} f(2) &= (2)^3 - 3(2)^2 + 1 \\ &= 8 - 12 + 1 \\ &= -3 \end{aligned}$$

Step 2: The values of f at its endpoints are:

- At $x = -\frac{1}{2}$:

$$\begin{aligned} f(-\frac{1}{2}) &= (-\frac{1}{2})^3 - 3(-\frac{1}{2})^2 + 1 \\ &= -\frac{1}{8} - \frac{3}{4} + 1 \\ &= \frac{1}{8} \end{aligned}$$

- At $x = 4$:

$$\begin{aligned} f(4) &= (4)^3 - 3(4)^2 + 1 \\ &= 64 - 48 + 1 \\ &= 17 \end{aligned}$$

Step 3: Comparing all these values:

- The absolute minimum of f is -3 attained at $x = 2$.
- The absolute maximum of f is 17 attained at $x = 4$.

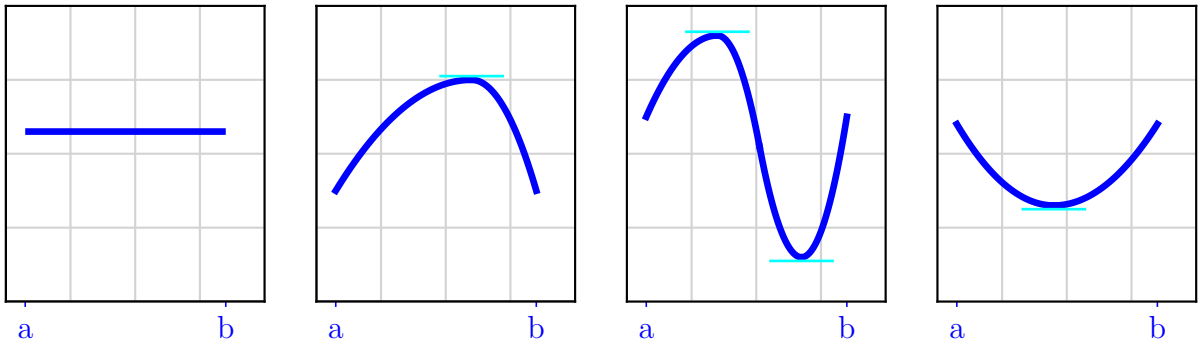
The Mean Value Theorem

To understand the Mean Value Theorem, we must first understand Rolle's Theorem:

- **Rolle's Theorem**:
  If f is a function that satisfies:
  - f is continuous on the closed interval $[a, b]$.
  - f is differentiable on the open interval (a, b) – everywhere except the endpoints.
  - $f(a) = f(b)$

  Then, there is a number $c \in (a, b)$ such that $f'(c) = 0$.

  Some examples:



- **Mean Value Theorem**:
  If f is a function that satisfies:
  - f is continuous on the closed interval $[a, b]$.
  - f is differentiable on the open interval (a, b).

  Then, there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

  or, $f(b) - f(a) = f'(c)(b - a)$.

- At some point, the average rate of change = instantaneous rate